Pratyush Rallapally

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EDUCATION

University of California, Santa Barbara **Sep 2023 - Mar 2027**
Majors: Statistics and Data Science, Biology (minors in Mathematics, French) GPA: 3.73/4.0 (Major GPA: 3.85/4.0)
Honors & Fellowships: College of Creative Studies Honors, Dean's Honors, Worster Undergraduate Fellow ($5000)

RESEARCH EXPERIENCE

Department of Ecology, Evolution, and Marine Biology, UC Santa Barbara **Jun 2025 - Present**
Undergraduate Researcher / Worster Undergraduate Fellow
Principal Investigator: Cherie Briggs, PhD

- Conducting independent research on longitudinal amphibian–parasite interaction networks to quantify temporal reassembly across ponds and years
- Applied network beta-diversity methods to partition temporal change into taxon-turnover and interaction-rewiring components, identifying turnover as the dominant source of annual network change
- Fit mixed-effects models and used cluster bootstrapping, and sensitivity analyses to assess the effects of sampling effort on pond-level temporal comparisons
- Integrated and cleaned ecological datasets in R and SQL and implemented a lab-wide relational database schema

San Luis Obispo County Probation Department, UCSB Data Science Capstone **Sep 2025 - May 2026**

- Conducted local validation of probation and pretrial risk-assessment tools using San Luis Obispo County administrative data
- Linked assessment and supervision records, constructed rearrest and reconviction outcomes, and developed separate validation pipelines across tools and supervision windows
- Evaluated predictive validity using logistic regression and ROC/AUC analysis, with stratified comparisons across tools, outcomes, and demographic groups

Department of Ecology, Evolution, and Marine Biology, UC Santa Barbara **Sep 2024 - May 2025**
Undergraduate Researcher
Principal Investigator: Cherie Briggs, PhD

- Developed a spatially explicit, individual-based model of invasive bullfrog population dynamics in Pleasanton Ridge Regional Park
- Parameterized life-history and movement processes using field census data, ecological literature, and GIS-derived habitat characteristics
- Constructed spatial inputs in QGIS and simulated individual movement and population trajectories to identify potential high-density population and disease hotspots
- Used model outputs to identify candidate locations for targeted invasive-species management in collaboration with East Bay Regional Park District

National Institute for Theory and Mathematics in Biology, Chicago, IL **Jun - Aug 2024**
NSF-funded Summer Undergraduate Research Program
Principal Investigator: Istvan Kovaćs, PhD

- Adapted the Hoshen–Kopelman algorithm in Python for two- and three-dimensional cluster analysis and large-scale simulations of gap-size distributions
- Ran large-scale simulations on high-performance computing clusters and applied resulting methods to Drosophila neuronal data in collaboration with Northwestern University neurobiology researchers

PUBLICATIONS

- **Rallapally, P.,** Ring, K., Briggs, C. J., et al. *Taxonomic turnover drives annual change in California amphibian–parasite networks.* Manuscript in preparation, 2026.

PRESENTATIONS

- Glass, C., **Rallapally, P.,** Sriram, A., Coburn, T., Heeren, L. **"Local Validation of Risk Assessment Tools in San Luis Obispo County Probation"** (2026). Central Coast Data Science Partnership Symposium, Santa Barbara, CA. *Poster Presentation*
- **Rallapally, P**., Ring, K. **"Dynamics of Host–Parasite Networks"** (2026). Worster Undergraduate Research Fellowship Symposium, UC Santa Barbara, CA. *Oral Presentation*
- **Rallapally, P.,** Ring, K., Briggs, C. J. (2025). "**Pixelating Persistence: A Spatially Explicit View of Bullfrog Dynamics"**
  Ecology, Evolution, and Marine Biology Undergraduate Research Symposium, UC Santa Barbara, CA. *Poster Presentation; Best Poster Award*
  UCSB Undergraduate Research Slam, UC Santa Barbara, CA. *Oral Presentation; Finalist*
  RACA-CON, College of Creative Studies, UC Santa Barbara, CA. *Poster Presentation*
- **Rallapally, P.**, Luzzatto, L. V., Kovács, I. A. **"Analysis and Applications of Gap-Size Statistics in 2D and 3D Clusters"** (2024). National Institute for Theory and Mathematics in Biology Summer Research Symposium, Northwestern University, Evanston, IL. *Poster Presentation*

WORK EXPERIENCE

Department of Mathematics, UC Santa Barbara **Sep 2026 - Present**
Undergraduate Learning Assistant

- Facilitate small-group problem solving for Math 34LA-LB; support first-year students with quantitative reasoning, study strategies, and navigation of the biology curriculum
- Adapt calculus labs into interactive Python notebooks to help students explore numerical methods and biological models

Department of Probability and Applied Statistics, UC Santa Barbara **Aug 2026 - Present**
Peer Advisor

- Advise undergraduate students on PSTAT majors, coursework, petitions, and degree requirements through email, drop-in advising, and one-on-one appointments
- Maintain course-equivalency and study-abroad databases for undergraduate advising

VOLUNTEER EXPERIENCE

Gauchos Vote Coalition - *Volunteer* **Jan 2025 – Present**

- Conduct nonpartisan voter-registration drives and outreach efforts across Santa Barbara County
- Organize tabling and informational events around voter-registration and election deadlines

SciTrek - *Science Educator* **Sep 2023 – Present**

- Lead hands-on science experiments, educational modules, and guided discussions for middle-school students

SKILLS & INTERESTS

- **Programming & Computing:** R, Python, SQL, SAS, MATLAB, C++, Bash/Git
- **Languages:** English (native), French (professional working proficiency)
- **Interests:** Juggling, Longboarding, Hiking, K-pop, Stand-Up/Sketch Comedy, Bouldering, Trivia